

Introduction to Multimodal Learning: Problem Set 1

Due: 3.28

1 Sampling, Aliasing, and DFT

1.1 Nyquist sampling

A continuous-time signal $x(t)$ is band-limited to angular frequency B , i.e. $X(j\omega) = 0$ for $|\omega| > B$.

1. What condition must the sampling angular frequency ω_s satisfy so that perfect reconstruction is possible? Explain in the frequency domain why aliasing happens when this condition is violated.
2. Why does applying a low-pass anti-aliasing filter before sampling help?

1.2 A concrete aliasing example

Consider the continuous-time sinusoid

$$x(t) = \cos(2\pi f_0 t).$$

Suppose it is sampled at sampling rate f_s , producing $x[n] = x(n/f_s)$.

1. Write $x[n]$ in terms of f_0 and f_s .
2. Let $f_0 = 9$ Hz and $f_s = 10$ Hz. Find a lower-frequency sinusoid that produces the same discrete-time samples. Explain why these two continuous-time signals become indistinguishable after sampling.

1.3 DFT, windowing, and FFT

Answer the following conceptual questions.

1. Why is the DFT more practical on a computer than the DTFT?
 2. In the lecture, truncating a signal for DFT is explained as multiplying by a rectangular window. Why can this hurt the frequency-domain result?
 3. Why can a Hann window produce a better spectrum than a rectangular window in many cases?
 4. Compare the computational complexity of direct DFT and FFT for an N -point signal.
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2 Image Compression Fundamentals

Briefly answer the following short questions.

2.1 DCT vs. DFT

Why is DCT typically preferred over DFT in practical image compression? Give at least **two** reasons based on the lecture.

2.2 Quantization

1. What is the role of quantization in the compression pipeline?
2. Why are high-frequency coefficients often quantized more aggressively than low-frequency coefficients?

2.3 Codeword assignment

1. What is the key idea behind variable-length coding?
2. What property must a variable-length code satisfy so that decoding is unambiguous?
3. Why does Huffman coding usually assign shorter codewords to more frequent symbols?

2.4 JPEG color space

JPEG commonly converts RGB images into YCbCr / YUV-like coordinates before compression.

1. Why is this transformation useful?
 2. Which channels can usually be compressed more aggressively, and why?
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3 Programming: Build a Mini JPEG-Style Compressor

In this problem, you will implement a simplified JPEG-style image compression pipeline using the provided images:

- p3_1.png
- p3_2.png

You may use `numpy`, `scipy`, `matplotlib`, `PIL`, and other standard Python libraries.

3.1 Patch-wise DCT

Implement block-based image transform on non-overlapping 8×8 patches.

1. Convert the image to grayscale first.
2. Apply 2D DCT to each 8×8 patch.
3. Visualize several example patches before and after DCT.
4. Briefly comment on where most of the energy tends to concentrate in the DCT coefficients.

3.2 Quantization design

Implement and compare the following two quantization strategies.

(A) Simple frequency-dependent quantization

Use

$$Q(i, j) = p \cdot (i + j + 1), \quad 0 \leq i, j \leq 7$$

with $p \in \{4, 8, 16\}$.

(B) JPEG luminance quantization matrix

Use the JPEG luminance quantization matrix from the lecture:

$$Q_Y = \begin{bmatrix} 16 & 11 & 10 & 16 & 24 & 40 & 51 & 61 \\ 12 & 12 & 14 & 19 & 26 & 58 & 60 & 55 \\ 14 & 13 & 16 & 24 & 40 & 57 & 69 & 56 \\ 14 & 17 & 22 & 29 & 51 & 87 & 80 & 62 \\ 18 & 22 & 37 & 56 & 68 & 109 & 103 & 77 \\ 24 & 35 & 55 & 64 & 81 & 104 & 113 & 92 \\ 49 & 64 & 78 & 87 & 103 & 121 & 120 & 101 \\ 72 & 92 & 95 & 98 & 112 & 100 & 103 & 99 \end{bmatrix}$$

and test scale factors $p \in \{0.5, 1, 2\}$, i.e. use pQ_Y .

For each strategy:

1. Quantize DCT coefficients by dividing by the quantization matrix and rounding to integers.
2. Reconstruct the image by dequantization and inverse DCT.
3. Display the original and reconstructed images side by side.

3.3 Compression statistics

For each image and each quantization setting, report:

1. Overall zero ratio of quantized coefficients.
2. Zero ratio in the low-frequency region:

$$\{(i, j) \mid i + j < 4\}$$

3. Zero ratio in the high-frequency region: all remaining coefficients.
4. PSNR between the original and reconstructed image.

Present your results in tables.

3.4 Zigzag scan and run-length encoding

1. Apply zigzag scan to each quantized 8×8 block.
2. Use run-length encoding (RLE) on the scanned coefficients.
3. Report the total encoded sequence length for each setting.
4. Compare the trend between zero ratio and RLE efficiency.

3.5 Analysis

Answer the following based on your experiments.

1. How does increasing the quantization scale affect visual quality, zero ratio, PSNR, and encoded length?
2. Which quantization strategy performs better overall, the simple hand-crafted matrix or the JPEG luminance matrix?
3. Does a higher zero ratio always imply better perceptual quality? Why or why not?
4. Which image is easier to compress, `p3_1.png` or `p3_2.png`? Explain based on image content.

Please submit your code and results.